Quantifier rules yield provability in tableaux

The Open Logic Project contributors

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-first-order-logic-tableaux/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the statement and proof use the tagenumerate/tagitem structure with the FOL tag active, so the full branch renders; the trees render as genuine tableaux under prooftrees

Editorial note. The label prop:provability-quantifiers names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own prop environment in the unit source.

Allowed standard prerequisites (editorial).

- Signed tableau rules for quantifiers. From False (exists x A(x)) the False-exists rule derives False A(t); from True (forall x A(x)) the True-forall rule derives True A(t).. Item (1) closes via the False-exists derivation against True A(t); item (2) closes via the True-forall derivation against False A(t).

Proposition 0.1. *1.* $A(t) \vdash \exists x\, A(x)$.

2. $\forall x\, A(x) \vdash A(t)$.

Proof. 1. A closed tableau for $\bot(\exists x\, A(x)), \top(A(t))$ is:

1.	$\bot(\exists x\, A(x))$	Ass
2.	$\top(A(t))$	Ass
3.	$\bot(A(t))$	$\bot\exists$[1]
	$\otimes$	

2. A closed tableau for $\bot(A(t)), \top(\forall x\, A(x))$, is:

1.	$\bot(A(t))$	Ass
2.	$\top(\forall x\, A(x))$	Ass
3.	$\top(A(t))$	$\top\forall$[2]
	$\otimes$	

□